Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Time-Series Forecasting

Time-series forecasting stands as a cornerstone of modern predictive analytics, vital for interpreting and anticipating the dynamic, state-dependent behaviors inherent to complex systems. Whether predicting fluctuating loads in renewable energy grids, identifying macro-economic trends, or monitoring structural health in industrial machinery, the ability to project historical sequential data into future horizons shifts

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.1 Pro, Gemini 3.5 Flash, Sonnet 4.6, and Opus 4.7, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

organizational strategy from reactive mitigation to proactive optimization. As systems grow increasingly interconnected and generate data at unprecedented scales, the demand for forecasting frameworks capable of parsing intricate temporal dynamics has driven a massive evolution in modeling methodologies.

Signal Acquisition, Pre-Processing, and the Nyquist-Shannon Theorem

Regardless of the downstream modeling architecture utilized, the predictive fidelity of any forecasting pipeline remains strictly bounded by the digital representation of the underlying signal. This representation relies heavily on initial data acquisition, signal conditioning, and pre-processing protocols.

The fundamental mathematical boundary governing the digitization of continuous physical processes is the Nyquist-Shannon sampling theorem. The theorem dictates that a continuous-time signal containing frequency components can be perfectly captured and reconstructed without loss of information if and only if it is sampled uniformly at a sampling frequency f_s that is strictly greater than twice the highest frequency component ($f_{\max}$) present within the signal:

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = \frac{f_s}{2}$ is defined as the Nyquist frequency. When a system violates this criterion—either due to an insufficient sampling rate or the absence of a proper low-pass anti-aliasing filter—high-frequency components beyond the Nyquist limit fold back into the lower spectrum. This distortive phenomenon, known as *aliasing*, introduces spurious, artificial low-frequency components that permanently mask and corrupt the true underlying dynamics of the signal. While classical aliasing is a hardware-level digitization artifact, an analogous form of structural distortion can manifest downstream within digital downsampling or patch-based tokenization operations, directly compromising a model's capacity to observe authentic temporal patterns.

Evolution of Forecasting Methodologies

Traditional Statistical Frameworks Historically, time-series analysis relied on classical statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). These traditional methods are deeply rooted in mathematical theory, offering high interpretability, exact statistical guarantees, and robust performance on stable, low-dimensional datasets. However, they rely heavily on strict underlying assumptions such as linearity, stationarity, and explicit seasonal decomposition. Consequently, classical approaches frequently struggle when confronted with high-frequency tracking, non-linear multi-rate variations, or large-scale datasets spanning cross-correlated variables.

Deep Learning and Transformer-Based Architectures To capture complex, non-linear temporal behaviors without rigid feature engineering, the paradigm shifted toward deep learning. Early architectures leveraged Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) to map temporal sequences sequentially.

More recently, the state of the art has been dominated by transformer-based architectures. By replacing recurrent bottlenecks with self-attention mechanisms, transformers can process sequences in parallel and explicitly compute dependencies between distant time steps. This allows the model to learn both localized high-frequency fluctuations and long-range macro-patterns simultaneously, forming highly adaptive architectures that can be fine-tuned to specific downstream industrial or scientific applications.

Time-Series Foundation Models and the Chronos Family Representing the cutting edge of this evolution is the emergence of time-series foundation models, most notably exemplified by the *Chronos* family of frameworks. Unifying time-series forecasting with the architectures powering modern natural language processing, Chronos treats time-series values not as continuous regressions, but as discrete language tokens.

The framework tokenizes real-valued historical data by quantizing values into discrete buckets via a scaling and binning operator. These numerical tokens are then fed directly into standard transformer language model backbones (such as T5). Trained on billions of diverse data points spanning synthetic and real-world domains, Chronos exhibits powerful zero-shot generalization, accurately predicting future trajectories across entirely unseen systems without requiring task-specific structural retraining or structural fine-tuning. This reliance on discrete tokenization windows, however, introduces novel architectural interactions with signal frequencies, exposing the model to localized structural vulnerabilities when processing periodic inputs.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0,\dots,P-1\}. \tag{3}$$

Generalized Mathematical Framework of Patch-Stride Phase-Locking

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi\frac{f}{f_s}n + \phi\right), \tag{4}$$

where f denotes the signal frequency satisfying the Nyquist criterion

$$f < \frac{f_s}{2}, \tag{5}$$

and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Derivation of the Phase-Locking Frequency Class Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S.

Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi\frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi\frac{f}{f_s}n + 2\pi\frac{fS}{f_s} + \phi\right). \tag{6}$$

For exact alignment with the tokenization grid, the phase accumulation over one stride must correspond to an integer number of full rotations:

$$2\pi\frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{7}$$

Solving for f yields the phase-locking frequency class

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S},\ c \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{8}$$

Defining the fundamental grid frequency

$$f_{\text{grid}} = \frac{f_s}{S}, \tag{9}$$

the admissible phase-locking frequencies correspond to integer harmonics of f_{grid} that remain below the Nyquist limit.

General Patch Indistinguishability Theorem

The collapse of consecutive tokens is not restricted to sinusoidal signals. Consider an arbitrary discrete-time signal satisfying the translational invariance condition

$$x[n+S] = x[n], \qquad \forall n. \tag{10}$$

Using Equation (3),

$$\mathbf{t}_{k+1}[m] = x[(k+1)S+m] = x[kS+m+S]. \tag{11}$$

Applying Equation (10) gives

$$\mathbf{t}_{k+1}[m] = x[kS+m] = \mathbf{t}_k[m]. \tag{12}$$

Since this equality holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{13}$$

Consequently, any signal exhibiting exact stride-period translational invariance produces identical consecutive tokens regardless of the patch length P.

Sinusoidal Signals as a Special Case

We now show that sinusoidal signals belong to the broader class defined above whenever their frequency lies in $\mathcal{F}_{\text{lock}}$.

Defining the normalized angular frequency

$$\omega_0 = 2\pi \frac{f}{f_s}, \tag{14}$$

and substituting Equation (8) yields

$$\omega_0 = \frac{2\pi c}{S} \quad \Longrightarrow \quad \omega_0 S = 2\pi c. \tag{15}$$

For any sample index n,

$$\begin{aligned} x[n+S] &= \sin(\omega_0(n+S)+\phi) && (16)\\ &= \sin(\omega_0 n + \omega_0 S + \phi) && (17)\\ &= \sin(\omega_0 n + 2\pi c + \phi) && (18)\\ &= \sin(\omega_0 n + \phi) && (19)\\ &= x[n]. && (20) \end{aligned}$$

Therefore every sinusoid satisfying

$$f = c\frac{f_s}{S} \tag{21}$$

belongs to the translationally invariant class of Equation (10), and thus satisfies

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{22}$$

Characterization of the Loss of Observability

When consecutive patches become indistinguishable, a localized structural aliasing event occurs. This degradation can be characterized through three complementary perspectives.

1. **Rank Deficiency in the Embedding Space**
   Let
   $$\mathbf{M} = [\mathbf{t}_k, \mathbf{t}_{k+1}, \ldots, \mathbf{t}_{k+L-1}] \in \mathbb{R}^{P\times L} \tag{23}$$
   be the local trajectory matrix.
   If two consecutive tokens become identical, then the corresponding columns become linearly dependent. Consequently, the trajectory matrix cannot achieve full rank:
   $$\operatorname{rank}(\mathbf{M}) < \min(P, L). \tag{24}$$
   More generally, every duplicated token introduces at least one linear dependency and therefore cannot increase the matrix rank. Repeated patch collapse systematically compresses the effective dimensionality of the local trajectory manifold and reduces the geometric diversity available to downstream embedding layers.
2. **Content Degeneration in Self-Attention**
   If consecutive patches satisfy
   $$\mathbf{t}_k = \mathbf{t}_{k+1}, \tag{25}$$
   their content representations become identical.
   For transformer architectures without positional information,
   $$\mathbf{q}_k = \mathbf{q}_{k+1}, \qquad \mathbf{k}_k = \mathbf{k}_{k+1}. \tag{26}$$
   Consequently, rows k and $k+1$ of the content-derived attention score matrix become identical.
   Modern transformer architectures generally employ positional encodings, rotary embeddings, or relative positional biases. Such mechanisms can still distinguish consecutive positions. Nevertheless, the content-derived component of the attention mechanism becomes degenerate, forcing the model to rely increasingly on positional information to break the symmetry introduced by token collapse.
3. **Loss of Sub-Stride Observability**
   The tokenization operator observes the signal only at stride-separated locations. This induces an effective observation frequency
   $$f_{\text{eff}} = \frac{f_s}{S}, \tag{27}$$
   with associated effective Nyquist frequency
   $$f_{N,\text{eff}} = \frac{f_s}{2S}. \tag{28}$$
   Signal dynamics occurring at temporal scales shorter than the stride interval become progressively harder to observe through the token sequence.
   Under phase-locking conditions, consecutive tokens become identical and the local representation collapses into a stationary state. Fine-grained phase evolution and high-frequency variations occurring

between token boundaries therefore become hidden from the observable representation presented to the model.

Analysis of Structural Degeneracies and Invariances

When $f \in \mathcal{F}_{\text{lock}}$, the resulting behavior depends on the relationship between P and S.

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping,

$$P = S. \tag{29}$$

Equation (10) directly implies

$$\mathbf{t}_{j+1} = \mathbf{t}_j \qquad \forall j. \tag{30}$$

The token sequence collapses into a static series of identical vectors, eliminating all observable temporal variation.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length,

$$S < P, \tag{31}$$

the translational invariance condition remains valid:

$$x[n+S] = x[n]. \tag{32}$$

Therefore,

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \tag{33}$$

Even though each token spans a larger temporal context, exact stride-period alignment forces consecutive tokens to remain identical. Consequently, overlap alone cannot prevent token-space degeneracy.

Case 3: Disjoint Windows ($S > P$) When

$$S > P, \tag{34}$$

gaps of length $S - P$ appear between successive windows.

Under the phase-locking condition,

$$x[n+S] = x[n], \tag{35}$$

which still guarantees

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \tag{36}$$

However, the interval separating consecutive windows remains completely unobserved by the tokenization operator. Consequently, distinct signals may produce exactly the same token sequence despite exhibiting different dynamics within the hidden intervals.

In this regime, the observation operator becomes non-injective: multiple underlying trajectories can map to an identical token representation. This creates a stronger form of information loss than simple token degeneracy because portions of the signal become permanently unrecoverable from the tokenized representation.

Domain Description and Ontology Representation

Structural Aliasing Expected Impact

Formalization of Falsifiable Hypotheses

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate f_s (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size P (`pv:patchSize`) and translation stride S (`pv:stride`).

The architectural grid frequency f_p (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \tag{37}$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

Hypothesis 1 (H_1): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules `R18` and `R19`).
- **Statement:** When the fundamental frequency f (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency f_p, the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

$$f = n \cdot f_p = n\left(\frac{f_s}{S}\right), \quad n \in \mathbb{N}^+ \tag{38}$$

  Under this boundary condition, the column rank of the local trajectory matrix collapses to unity, rendering the internal dynamics of the patch unobservable to downstream self-attention mechanisms.
- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band $[2, \frac{f_s}{2}]$ Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of f_p.

Hypothesis 2 (H_2): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule `R20`).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal's peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency $f \in \mathcal{F}_{\text{lock}}$, translating the initial phase offset ϕ (`pv:phaseRad`) within the discrete-time sinusoid:

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \quad \phi \in [0, 2\pi) \tag{39}$$

  modifies the specific sub-sequence frozen inside each individual patch window. If the phase shift satisfies $\phi \in (0, S)$, it induces a structural misalignment that alters the component-wise values of consecutive token vectors, systematically modulating the token space fidelity.

- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric SV) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements $\phi \sim \text{Uniform}(0, 2\pi)$ at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

Hypothesis 3 (H_3): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule `R21`).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.
- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio or (`pv:overlapRatio`):

$$or = \frac{P - S}{P} \tag{40}$$

  When $or = 0$ ($P = S$), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ($\mathbf{t}_j = \mathbf{t}_{j+1}$), wiping out temporal gradients. When $or \to 1$ ($S \ll P$), successive patches share massive historical context; although the stride-bound phase-locking condition $x[n + S] = x[n]$ still holds, the expanded patch dimension P forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.
- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters P and S while keeping a locked signal frequency f yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

Signal Generation

Simple sinusoid generation. TSMixup and KernelSynth description, algorithm pseudo-code.